LAB REPORT: ASSIGNMENT 02

Topic: Creating New Columns Using Python & R

- **Course Name:** Introduction To Data Science
- **Course Code :** ADS 201
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1. Objective

The objective of this assignment is to create new columns in a dataset and apply conditional logic using both Python and R. The assignment demonstrates how to generate calculated values and classify records based on a specified condition.

2. Dataset Overview

- **Dataset Name:** Space_Corrected.csv
- **Description:** The dataset contains historical information about space mission launches, including company names, launch locations, mission details, rocket status, mission status, and other related information. For this assignment, additional practice columns were created to perform calculations and apply conditional operations.

3. Part A: Implementation in Python

Steps Taken:

1. Loaded the dataset using the pandas library.
2. Created two new numeric columns named Score1 and Score2.
3. Calculated the average of the two scores and stored it in a new column named Average_Score.
4. Applied a condition to create a new Status column. If the average score was 85 or above, the status was set to "Pass"; otherwise, it was set to "Fail".
5. Displayed the newly created columns to verify the results..

Python Code:

```python
import pandas as pd

# Load dataset
df = pd.read_csv("Space_Corrected.csv")

# 1. Create dummy score columns
df["Score1"] = (df.index % 21) + 70      # 70-90
df["Score2"] = (df.index % 16) + 75      # 75-90

# 2. Create 'Average Score' column
df["Average Score"] = df[["Score1", "Score2"]].mean(axis=1)

# 3. Apply condition
df["Status"] = df["Average Score"].apply(
    lambda x : "Pass" if x >= 85 else "Fail"
)

# Display output
print(df[["Score1", "Score2", "Average Score", "Status"]].head())
```

Output & Observations:

The Python program successfully created four new columns: Score1, Score2, Average_Score, and Status. The average score was calculated correctly, and each record was classified as Pass or Fail based on the specified condition.

```
   Score1  Score2  Average Score Status
0      70      75           72.5   Fail
1      71      76           73.5   Fail
2      72      77           74.5   Fail
3      73      78           75.5   Fail
4      74      79           76.5   Fail

Process finished with exit code 0
```

4. Part B: Implementation in R

Steps Taken:

1. Loaded the dataset using the read.csv() function.
2. Created two new numeric columns named Score1 and Score2.
3. Calculated the average score and stored it in the Average_Score column.
4. Applied conditional logic using ifelse() to create the Status column.
5. Displayed the generated columns to verify the output.

R Code:

```
assignment_1.R* ×   Untitled1 ×   assignment_2.py ×   assignment_2.R* ×
Source on Save
# Load dataset
df <- read.csv("Space_Corrected.csv")

# 1. Create dummy score columns
df$Score1 <- (seq_len(nrow(df)) - 1) %% 21 + 70
df$Score2 <- (seq_len(nrow(df)) - 1) %% 16 + 75

# 2. Create 'Average Score' column
df$Average_Score <- (df$Score1 + df$Score2) / 2

# 3. Apply condition
df$Status <- ifelse(df$Average_Score >= 85, "Pass", "Fail")

# Display output
print(head(df[c("Score1", "Score2", "Average_Score", "Status")]))
```

Output & Observations:

The R program successfully generated the required columns and produced the same results as the Python implementation. The average score was calculated correctly, and the records were classified according to the specified condition.

```
Console   Terminal ×   Background Jobs ×
R 4.6.1 · ~/PycharmProjects/Data_Analysis_Assignments1/
> # Load dataset
> df <- read.csv("Space_Corrected.csv")
>
> # 1. Create dummy score columns
> df$Score1 <- (seq_len(nrow(df)) - 1) %% 21 + 70
> df$Score2 <- (seq_len(nrow(df)) - 1) %% 16 + 75
>
> # 2. Create 'Average Score' column
> df$Average_Score <- (df$Score1 + df$Score2) / 2
>
> # 3. Apply condition
> df$Status <- ifelse(df$Average_Score >= 85, "Pass", "Fail")
>
> # Display output
```

```
R 4.6.1 · ~/PycharmProjects/Data_Analysis_Assignments1/
> # 3. Apply condition
> df$Status <- ifelse(df$Average_Score >= 85, "Pass", "Fail")
>
> # Display output
> print(head(df[c("Score1", "Score2", "Average_Score", "Status")]))
  Score1 Score2 Average_Score Status
1     70     75          72.5   Fail
2     71     76          73.5   Fail
3     72     77          74.5   Fail
4     73     78          75.5   Fail
5     74     79          76.5   Fail
6     75     80          77.5   Fail
>
```

5. Conclusion

Both Python and R successfully completed the required tasks for this assignment. New columns were created, the average score was calculated correctly, and conditional logic was applied to classify each record. The results produced by both programming languages were consistent, demonstrating that the same data manipulation task can be performed effectively in either environment.